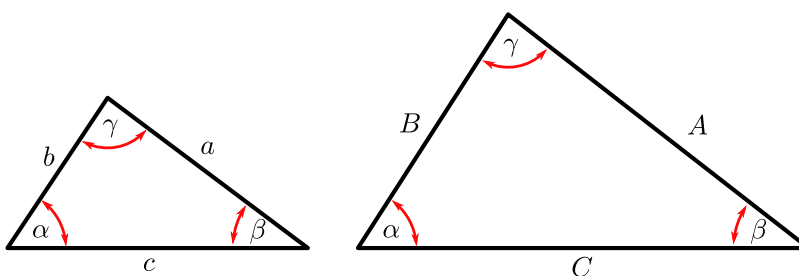


HIGH SCHOOL MATERIAL

This chapter is really split into two parts.

- Sections [A.1](#) to [A.13](#) we expect you to understand and know.
- The very last section, Section [A.14](#), contains results that we don't expect you to memorise, but that we think you should be able to quickly derive from other results you know.

A.1 ▴ Similar Triangles



Two triangles T_1, T_2 are similar when

- (AAA — angle angle angle) The angles of T_1 are the same as the angles of T_2 .
- (SSS — side side side) The ratios of the side lengths are the same. That is

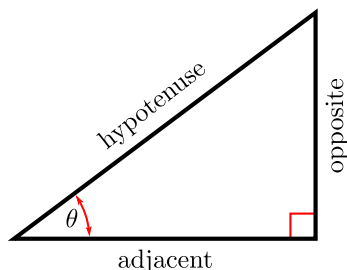
$$\frac{A}{a} = \frac{B}{b} = \frac{C}{c}$$

- (SAS — side angle side) Two sides have lengths in the same ratio and the angle between them is the same. For example

$$\frac{A}{a} = \frac{C}{c} \text{ and angle } \beta \text{ is same}$$

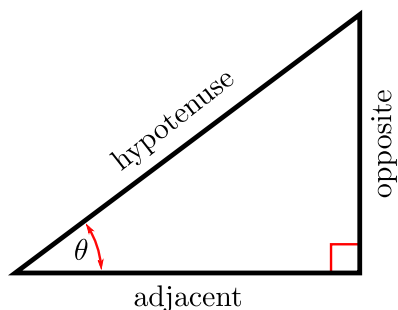
A.2 ▲ Pythagoras

For a right-angled triangle the length of the hypotenuse is related to the lengths of the other two sides by



$$(\text{adjacent})^2 + (\text{opposite})^2 = (\text{hypotenuse})^2$$

A.3 ▲ Trigonometry — Definitions



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

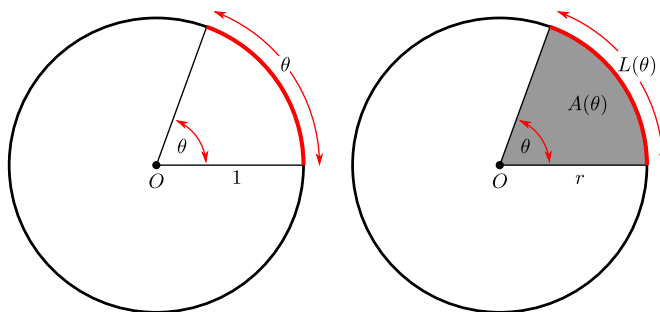
$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

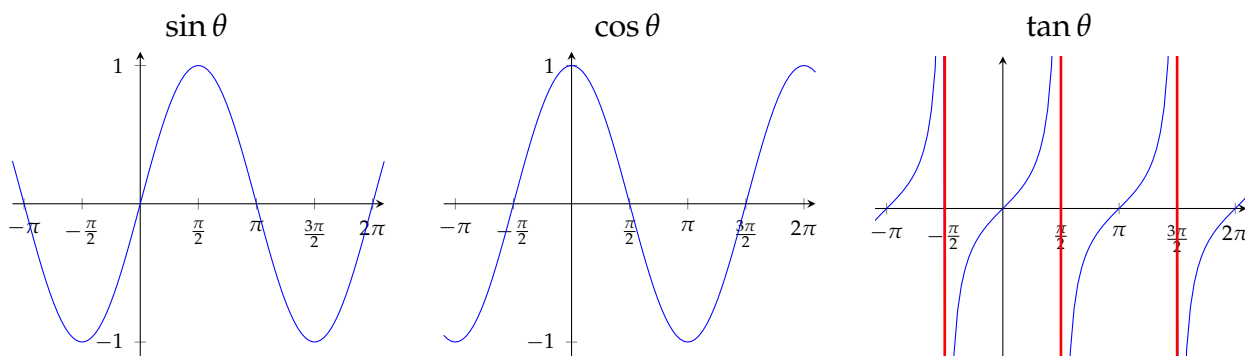
A.4 ▲ Radians, Arcs and Sectors



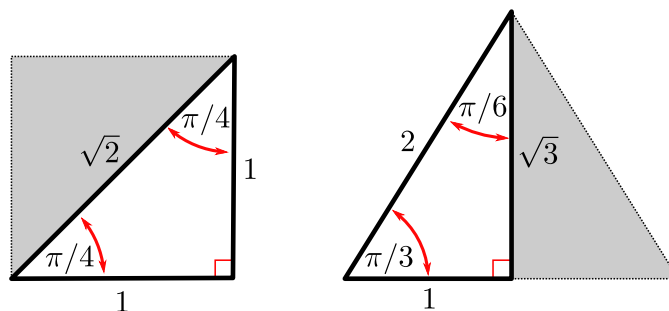
For a circle of radius r and angle of θ radians:

- Arc length $L(\theta) = r\theta$.
- Area of sector $A(\theta) = \frac{\theta}{2}r^2$.

A.5 ▴ Trigonometry — Graphs



A.6 ▴ Trigonometry — Special Triangles



From the above pair of special triangles we have

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\tan \frac{\pi}{4} = 1$$

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\tan \frac{\pi}{3} = \sqrt{3}$$

A.7 ▴ Trigonometry — Simple Identities

- Periodicity

$$\sin(\theta + 2\pi) = \sin(\theta)$$

$$\cos(\theta + 2\pi) = \cos(\theta)$$

- Reflection

$$\sin(-\theta) = -\sin(\theta)$$

$$\cos(-\theta) = \cos(\theta)$$

- Reflection around $\pi/4$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

- Reflection around $\pi/2$

$$\sin(\pi - \theta) = \sin \theta$$

$$\cos(\pi - \theta) = -\cos \theta$$

- Rotation by π

$$\sin(\theta + \pi) = -\sin \theta$$

$$\cos(\theta + \pi) = -\cos \theta$$

- Pythagoras

$$\sin^2 \theta + \cos^2 \theta = 1$$

A.8 ▲ Trigonometry — Add and Subtract Angles

- Sine

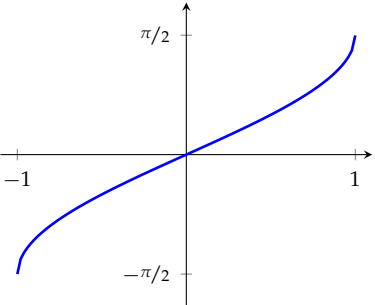
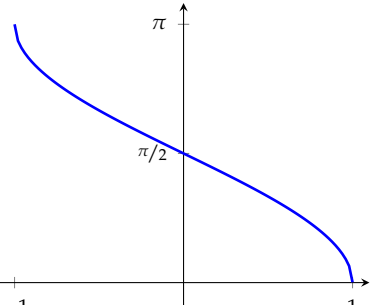
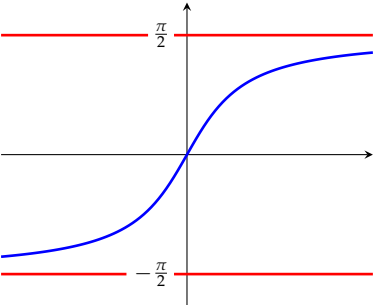
$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

- Cosine

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$$

A.9 ▲ Inverse Trigonometric Functions

Some of you may not have studied inverse trigonometric functions in highschool, however we still expect you to know them by the end of the course.

$\arcsin x$	$\arccos x$	$\arctan x$
Domain: $-1 \leq x \leq 1$ Range: $-\frac{\pi}{2} \leq \arcsin x \leq \frac{\pi}{2}$	Domain: $-1 \leq x \leq 1$ Range: $0 \leq \arccos x \leq \pi$	Domain: all real numbers Range: $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$
		

Since these functions are inverses of each other we have

$$\arcsin(\sin \theta) = \theta$$

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\arccos(\cos \theta) = \theta$$

$$0 \leq \theta \leq \pi$$

$$\arctan(\tan \theta) = \theta$$

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

and also

$$\sin(\arcsin x) = x$$

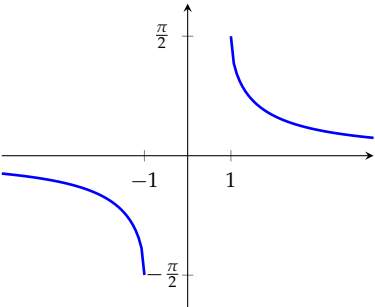
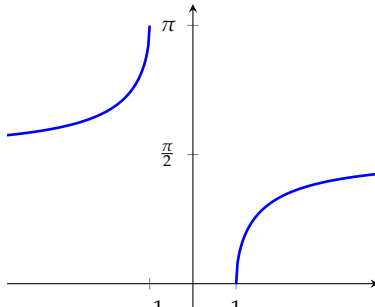
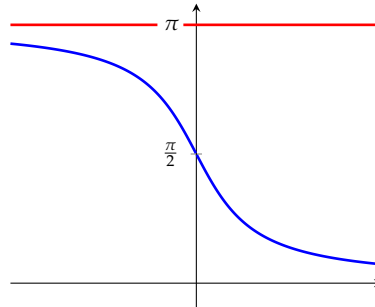
$$-1 \leq x \leq 1$$

$$\cos(\arccos x) = x$$

$$-1 \leq x \leq 1$$

$$\tan(\arctan x) = x$$

$$\text{any real } x$$

$\operatorname{arccsc} x$	$\operatorname{arcsec} x$	$\operatorname{arccot} x$
Domain: $ x \geq 1$ Range: $-\frac{\pi}{2} \leq \operatorname{arccsc} x \leq \frac{\pi}{2}$ $\operatorname{arccsc} x \neq 0$	Domain: $ x \geq 1$ Range: $0 \leq \operatorname{arcsec} x \leq \pi$ $\operatorname{arcsec} x \neq \frac{\pi}{2}$	Domain: all real numbers Range: $0 < \operatorname{arccot} x < \pi$
		

Again

$$\operatorname{arccsc}(\csc \theta) = \theta$$

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \theta \neq 0$$

$$\operatorname{arcsec}(\sec \theta) = \theta$$

$$0 \leq \theta \leq \pi, \theta \neq \frac{\pi}{2}$$

$$\operatorname{arccot}(\cot \theta) = \theta$$

$$0 < \theta < \pi$$

and

$$\csc(\operatorname{arccsc} x) = x$$

$$|x| \geq 1$$

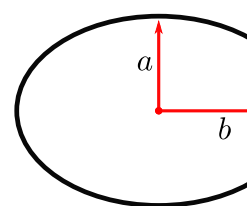
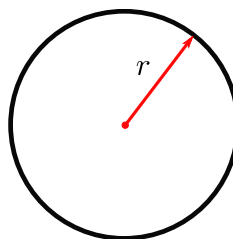
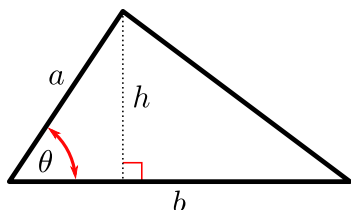
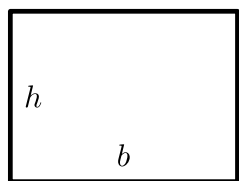
$$\sec(\operatorname{arcsec} x) = x$$

$$|x| \geq 1$$

$$\cot(\operatorname{arccot} x) = x$$

$$\text{any real } x$$

A.10 ▀ Areas



- Area of a rectangle

$$A = bh$$

- Area of a triangle

$$A = \frac{1}{2}bh = \frac{1}{2}ab \sin \theta$$

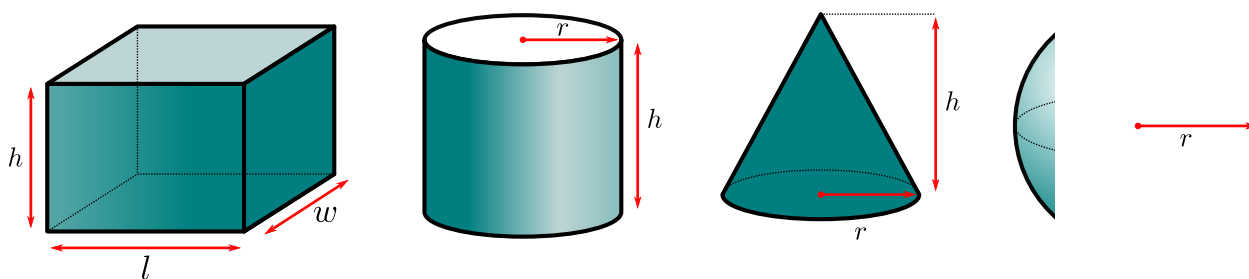
- Area of a circle

$$A = \pi r^2$$

- Area of an ellipse

$$A = \pi ab$$

A.11 ▲ Volumes



- Volume of a rectangular prism

$$V = lwh$$

- Volume of a cylinder

$$V = \pi r^2 h$$

- Volume of a cone

$$V = \frac{1}{3}\pi r^2 h$$

- Volume of a sphere

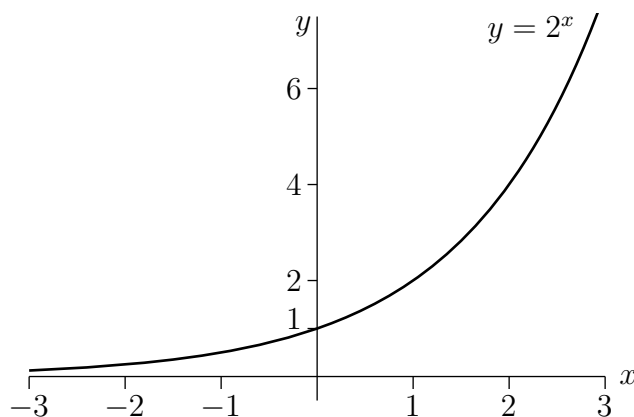
$$V = \frac{4}{3}\pi r^3$$

A.12 ▲ Powers

In the following, x and y are arbitrary real numbers, and q is an arbitrary constant that is strictly bigger than zero.

- $q^0 = 1$

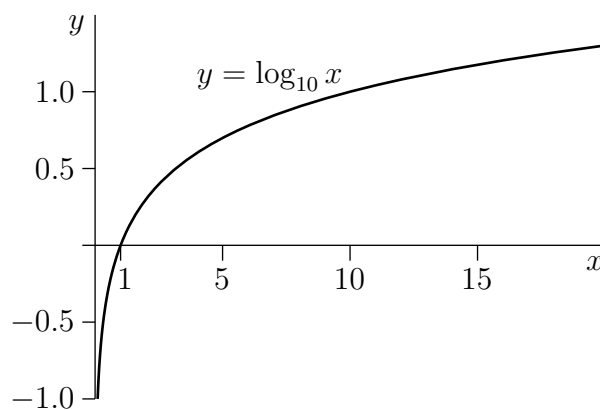
- $q^{x+y} = q^x q^y, q^{x-y} = \frac{q^x}{q^y}$
- $q^{-x} = \frac{1}{q^x}$
- $(q^x)^y = q^{xy}$
- $\lim_{x \rightarrow \infty} q^x = \infty, \lim_{x \rightarrow -\infty} q^x = 0$ if $q > 1$
- $\lim_{x \rightarrow \infty} q^x = 0, \lim_{x \rightarrow -\infty} q^x = \infty$ if $0 < q < 1$
- The graph of 2^x is given below. The graph of q^x , for any $q > 1$, is similar.



A.13 ▲ Logarithms

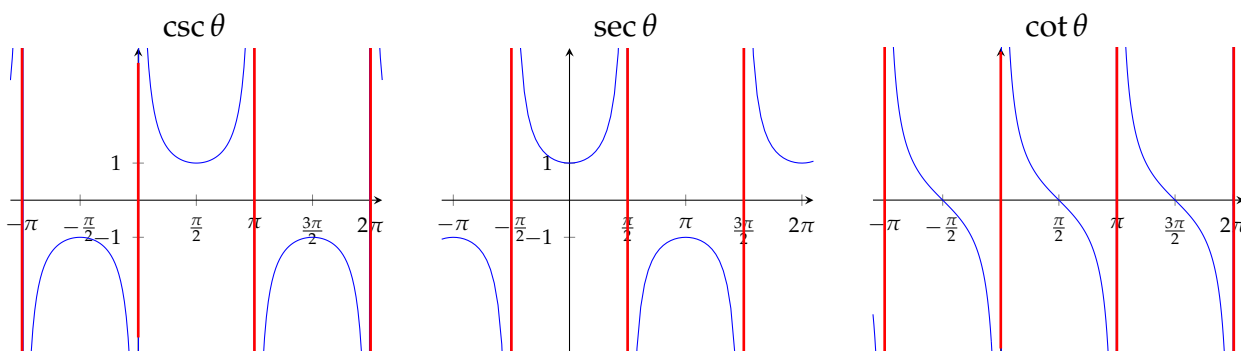
In the following, x and y are arbitrary real numbers that are strictly bigger than 0, and p and q are arbitrary constants that are strictly bigger than one.

- $q^{\log_q x} = x, \log_q (q^x) = x$
- $\log_q x = \frac{\log_p x}{\log_p q}$
- $\log_q 1 = 0, \log_q q = 1$
- $\log_q (xy) = \log_q x + \log_q y$
- $\log_q \left(\frac{x}{y}\right) = \log_q x - \log_q y$
- $\log_q \left(\frac{1}{y}\right) = -\log_q y,$
- $\log_q (x^y) = y \log_q x$
- $\lim_{x \rightarrow \infty} \log_q x = \infty, \lim_{x \rightarrow 0+} \log_q x = -\infty$
- The graph of $\log_{10} x$ is given below. The graph of $\log_q x$, for any $q > 1$, is similar.



A.14 ▲ Highschool Material You Should be Able to Derive

- Graphs of $\csc \theta$, $\sec \theta$ and $\cot \theta$:



- More Pythagoras

$$\begin{array}{lll} \sin^2 \theta + \cos^2 \theta = 1 & \xrightarrow{\text{divide by } \cos^2 \theta} & \tan^2 \theta + 1 = \sec^2 \theta \\ \sin^2 \theta + \cos^2 \theta = 1 & \xrightarrow{\text{divide by } \sin^2 \theta} & 1 + \cot^2 \theta = \csc^2 \theta \end{array}$$

- Sine — double angle (set $\beta = \alpha$ in sine angle addition formula)

$$\sin(2\alpha) = 2 \sin(\alpha) \cos(\alpha)$$

- Cosine — double angle (set $\beta = \alpha$ in cosine angle addition formula)

$$\begin{array}{ll} \cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha) & \\ = 2 \cos^2(\alpha) - 1 & \text{(use } \sin^2(\alpha) = 1 - \cos^2(\alpha) \text{)} \\ = 1 - 2 \sin^2(\alpha) & \text{(use } \cos^2(\alpha) = 1 - \sin^2(\alpha) \text{)} \end{array}$$

- Composition of trigonometric and inverse trigonometric functions:

$$\cos(\arcsin x) = \sqrt{1 - x^2} \qquad \sec(\arctan x) = \sqrt{1 + x^2}$$

and similar expressions.